

Lesson 8: Investigating I-V characteristics

Electric Circuits · Year 10 · 50 minutes · AQA RP 16 / RP 4

Why this lesson is here

Changing only the test component while keeping the measuring circuit clear lets you compare its electrical behaviour.

Learning objective

Collect bipolar I-V data safely, graph it and justify the three shapes.

Core teaching point

Vary p.d., record current, reverse the supply for negative readings. Keep a fixed resistor cool; protect the diode with a series resistor; do not exceed the lamp rating.

Equipment and preparation

Variable low-voltage supply, variable resistor, meters, lamp, fixed resistor, diode, protective resistor, switch, leads; existing I-V worksheet pp. 1-8.

Lesson sequence

Time	Phase	Teacher move and evidence
0-6	Safety and prediction	Review lamp rating and diode protection.
6-13	Build	Check series ammeter and parallel voltmeter.
13-34	Required practical	Collect all three devices and both directions; rotate roles.
34-44	Graph	Plot characteristic curves; finish if needed.
44-50	Check	Name and explain each shape.

Workbook task guidance

Task 1: Draw a series supply-switch-ammeter-variable resistor-test component loop and voltmeter across the test component.

Expected: Ammeter and variable resistor in series with test component; voltmeter across test component only.

Task 2: Collect readings for fixed resistor, lamp and protected diode, including reverse direction. Record negative signs correctly.

Expected: Student-dependent readings; negative V and I in reverse direction. Use diode protection and lamp rating.

Task 3: Plot I (y) against V (x) for each device and label the origin and units.

Expected: I vertical, V horizontal; put negative readings in the third quadrant.

Task 4: Explain the shapes with constant temperature, filament heating and one-way diode conduction.

Expected: Fixed resistor: constant R; lamp heats, R rises; diode mainly conducts forward.

Worked example

For a fixed resistor, $V = +3.0 \text{ V}$ and $I = +0.15 \text{ A}$ gives $R = 20 \text{ ohm}$. Reversed readings may be -3.0 V and -0.15 A .

AQA-style checkpoint

[6 marks] Describe a method to obtain the complete I-V characteristic of a filament lamp and state two ways the method changes for a diode.

Mark guidance: Award up to six: correct ammeter series and voltmeter parallel; vary p.d. with supply/variable resistor; record V and I pairs; reverse supply for negative half; keep below lamp rating; diode gets protective series resistor; smaller voltage

steps/milliammeter for diode; plot I against V .

Anticipated misconception

Reversing the supply changes both V and I signs; do not plot reversed data in the positive quadrant.

Assessment and next step

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.

Practical hand-off

Use the workbook practical record for data and graphing. The existing full AQA required-practical worksheet supplies detailed method, circuit diagrams and additional analysis. Use low voltage and local lab safety rules.